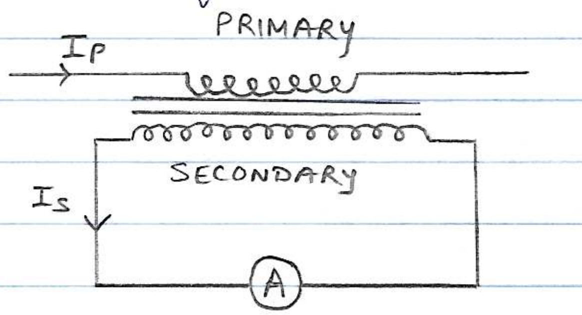
BY AKHIL K.

→ INSTRUMENT TRANSFORMERS

- The extension of instrument range, so that the curre and voltage can be measured with instrument of modera size and capacity is important. Range extension can be accomplished by ~~by~~ use of shunts for current and multipliers to voltage measurements, but this method is only practical for relatively small values of currents and voltages.

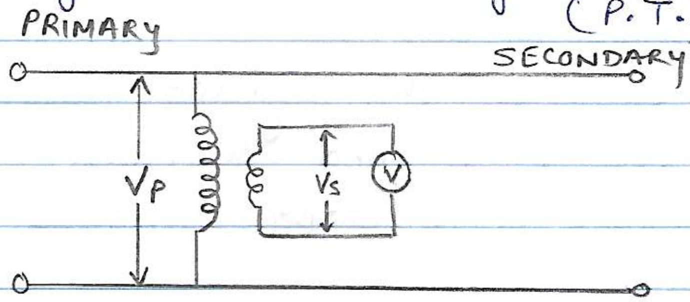
- What are instrument transformers?
  High voltages / currents have to be stepped down by transformers to values more convenient for measureme Transformers used in connection with instruments for measurement function are called instrument transfor

- The transformer used for measurement of current is called current transformer (C.T.) while transfor used for measurement of voltage is called voltage transfo (P.T.)



CURRENT TRANSFORMER

The primary winding is so connected that the current being measured passes through it and the secondary is connected to an ammeter. The current is stepped down to the current level of ammeter. C.T. is ~~also known as~~ a <u>step up transformer.</u>

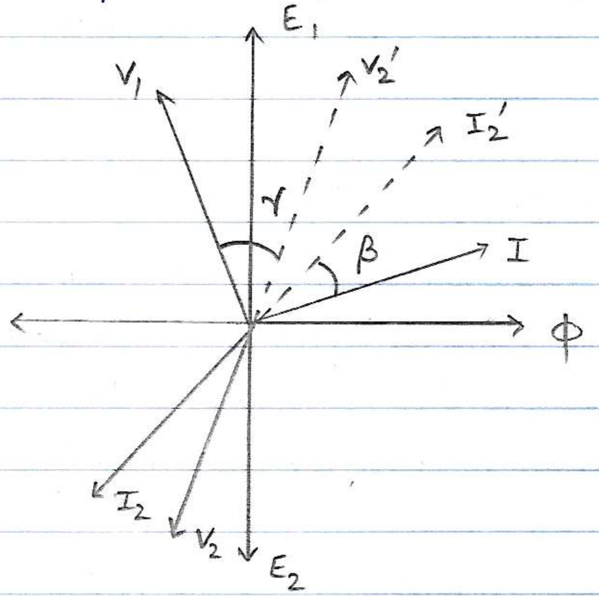


POTENTIAL TRANSFORMER

The primary is connected to the voltage being measured and the secondary to voltme The potential transformer steps down the voltage to the level of voltmeter. P.T. is a <u>step down transfo -er.</u>

RATIO AND PHASE ANGLE ERRORS

- In reality the transformation ratio doesn't remain constant in case of an instrument transformer. Instead it depends upon the exciting current and power factor of the secondary circuit. This is called ratio error which depends upon the working component of primary current.



- The phase angle between primary and secondary current is not exactly 180° but slightly less than this.

- The angular displacement β between I_1 and I_2' (I_2 reversed) is called phase angle error of current transformers. It is :

(*) +ve if reversed secondary current leads primary current
(*) on very low power factors it can be negative.

- Similarly the angular displacement γ between primary voltage V_1 and secondary reversed (V_2') is called phase angle error of a voltage transformer (P.T.).

- It depends upon the magnetising component of primary current.

Nominal transformation ratio: It is the ratio of rated primary to rated secondary, denoted by k_n.

$$k_n = \frac{\text{rated primary current } (I_1)}{\text{rated secondary current } (I_2)} \quad \text{for C.T.}$$

$$K_n = \frac{\text{rated primary voltage } (V_1)}{\text{rated secondary voltage } (V_2)} \quad \text{for P.T.}$$

Actual transformation ratio : The ~~got~~ actual transformation ratio under any loading condition.

$$k = \frac{\text{primary current } (I_1)}{\text{corresponding secondary current } (I_2)}$$

Ratio error : It is given by :

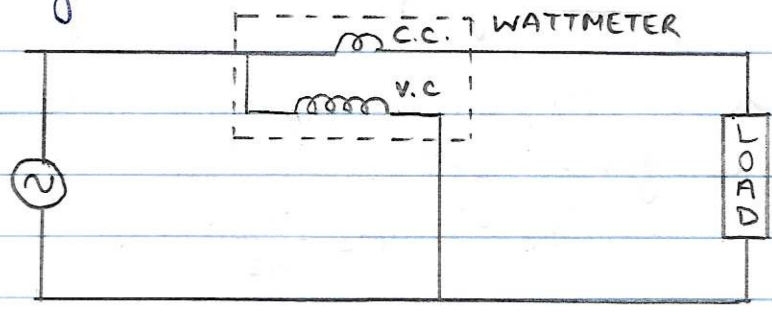
$$\text{Ratio error} = \frac{k_n - k}{k}.$$

MEASUREMENT OF POWER

- Power in d.c. circuit can be measured with the help of Voltmeter and ammeter. However in A.C. circuit the power is given by $P = VI\cos\phi$.
  where $\cos\phi$ is the power factor of the circuit which is defined as the ratio of true power to apparant power.
  It is due to the power factor we use a wattmeter for measurement of power in AC circuits because voltmeter - ammeter method takes no account of power factor.

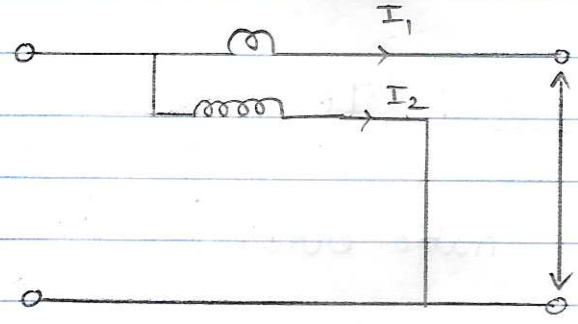
WATTMETERS

A wattmeter consists of a current coil (C.C.) which carries load current and a voltage coil (V.C or P.C.) which carries a current proportional to and in phase with the voltage as shown.



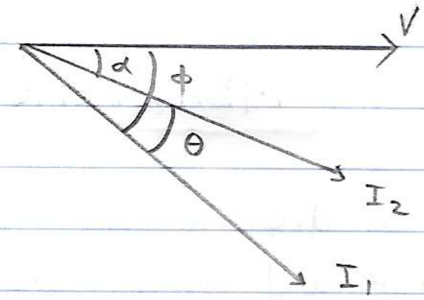
Error due to P.C. inductance

The wattmeter reading is proportional to $I_1 I_2 \cos\theta$ where θ is the angle between the two currents I_1 and I_2 as shown.



When we take the inductance of the P.C. into consideration, the wattmeter reading

$$\propto \frac{V}{R}\cos\alpha\cos(\phi-\alpha).$$



The correction factor is the ratio of true reading W_t (which is $\propto \frac{I_1 V}{R}\cos\phi$) and the actual reading W_a (which is $\propto \frac{V I_1}{R}\cos\alpha\cos(\phi-\alpha)$).

$$\text{Ratio} = \frac{\text{True power } (W_t)}{\text{Actual wattmeter reading } (W_a)}$$

$$= \frac{\cancel{\frac{VI_1}{R}}\cos\phi}{\cancel{\frac{VI_1}{R}}\cos\alpha\cos(\phi-\alpha)}$$

$$\boxed{\text{True power} = \frac{\cos\phi}{\cos\alpha\cos(\phi-\alpha)} \times \text{Actual reading (due to pressure coil inductance)}}$$

Expression for controlling and deflecting torque

The instantaneous torque of a dynamometer type instrument is given by:

$$T_i = i_1 i_2 \frac{dM}{d\theta}$$

where i_1 & i_2 are instantaneous values of currents in two pressure coils.

Instantaneous value of voltage across pressure coil circuit

$$v = \sqrt{2}\, V \sin \omega t$$

- Current i_p in P.C. is in phase with voltage as R_p is very

$$i_p = \frac{v}{R_p} = \frac{\sqrt{2}\, V}{R_p} \sin \omega t = \sqrt{2}\, I_p \sin \omega t$$

and $I_p = \frac{V_p}{R_p}$ (rms value of current in P.C.)

- If the current in C.C. lags the voltage by angle ϕ, inst. value of current through coil is

$$i_p = \sqrt{2}\, I \sin(\omega t - \phi)$$

Thus instantaneous torque $T_i = I_p I \left(\cos\phi - \cos(2\omega t - \phi) \right) \frac{dM}{d\theta}$

AVERAGE DEFLECTING TORQUE

$$T_d = \frac{1}{T} \int_0^T T_i \, d(\omega t)$$

$$\boxed{T_d = \frac{VI}{R_p} \cos\phi \frac{dM}{d\theta}}$$

CONTROLLING TORQUE

$T_c = k\theta = T_d$ (at balance position)

$$\boxed{\theta = \left(\frac{VI \cos\phi}{R_p K} \right) \frac{dM}{d\theta}}$$

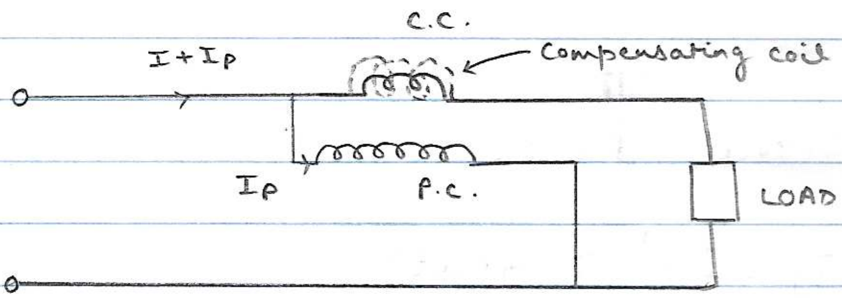
LOW POWER FACTOR WATTMETERS

Measurement of power in LPF circuits is difficult because:

1) deflecting torque is small
2) errors due to inductance of P.C. are large in LPF circuits.

Features of a LPF wattmeter:

1) The P.C. is designed to have a low value of resistance for increased operating torque. The I_P can be 10 times as that of conventional wattmeters.
2) A compensating winding may be used to compensate for the error caused by power loss in pressure coil circuit.



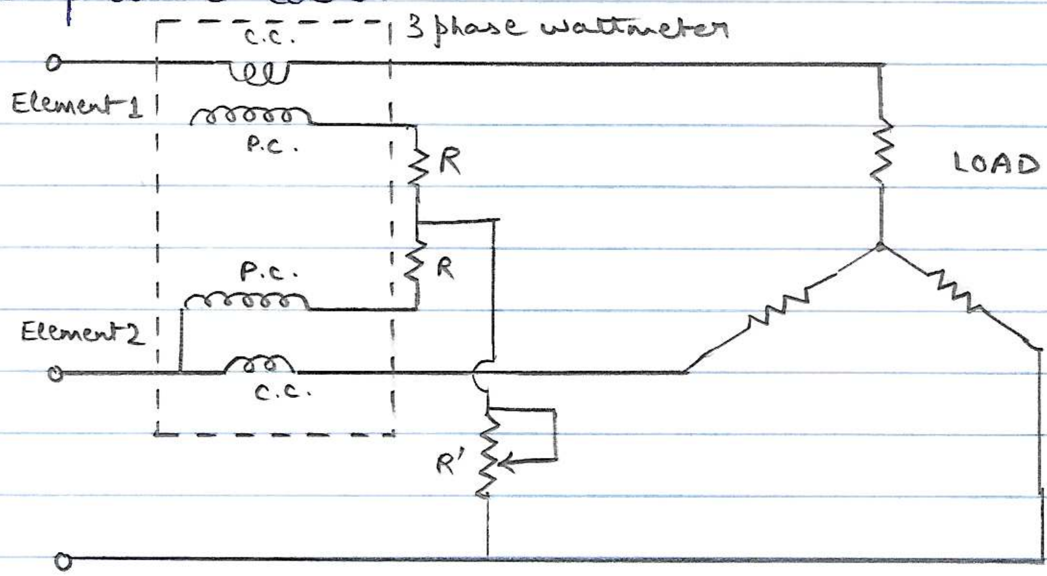
The compensating coil is connected in series with P.C. and opposes the field of the current coil. The compensating coil carries I_P current and produces a field which opposes the C.C. ~~coil~~ field. Thus resultant field is due to I only and error caused by pressure coil current flowing in C.C. is neutralized.

3) To compensate for error caused by inductance of P.C., a capacitor can also be connected across a part of series resistance in P.C. circuit.

4) LPF wattmeters can also be designed with small controlle torque so as to give full scale deflection for power factors as low as 0.1.

THREE PHASE WATTMETERS

They consist of 2 separate wattmeters movements mounted together in one case with two moving coils mounted on same spindle. There are two current coils and two pressure coils.



THREE PHASE WATTMETER

The torque on each element is proportional to power measured by it.

$\therefore$ Total deflecting torque $\propto (P_1 + P_2) \propto P_{total}$

In order for a 3 phase wattmeter to read correctly, there should not be any mutual interference between the two eleme Thus a laminated iron shild is used. Resistance R may be adjusted to compensate for errors caused by mutual interference.